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2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence

4
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9 **Abstract**

10 Advances in artificial intelligence (AI) have enabled the wide use of personalized
11 assistance for cognitive tasks, which enhanced individual performance on various
12 fields. However, the impact of AI assistance on collective intelligence remains poorly
13 understood. AI assistance may undermine the emergence of collective intelligence by
14 reducing the individual diversity or introducing structured biases into a population.
15 Here, we develop a human-AI collective prediction model mediated by social learning
16 based on incentive structure by extending the existing works on collective intelligence.
17 Our model uncovers that emergence of collective intelligence is depend on how human
18 and AI interact. When AI works as if it is omniscient and gives assistance regardless
19 of users' interests, collective intelligence will be promoted. However, when AI works
20 as a chat-bot and gives assistance only based on users' interests, collective intelligence
21 could be hindered due to diversity loss and overconfidence on AI. We showed that,
22 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
23 Instead, elaborating more diversity-aware incentives could be the key to promote and
24 maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

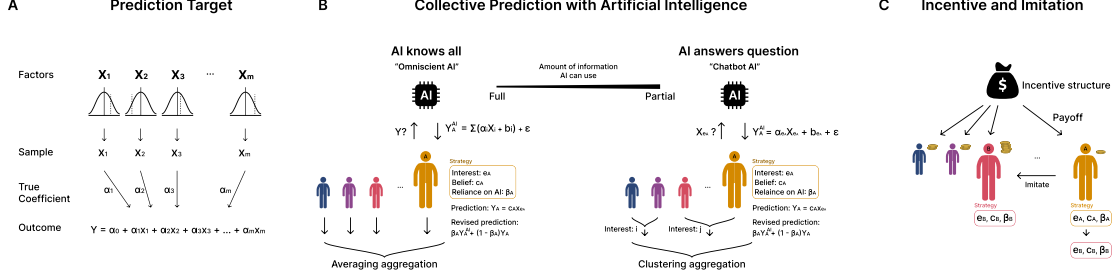


Figure 1: A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants. [SWP: Need brief description.]

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{eA} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{eA}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). This revised prediction
 79 is aggregated to form collective prediction. [SWP: I think we need to say averaging
 80 and clustering aggregation here?]

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless
 88 of the user's interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user's interest:
 90 $Y_A^{AI} = \alpha_{eA} x_{eA} + b_{eA} + \epsilon$. Hereafter, we refer to these two types as 'Omniscient AI'
 91 and 'Chatbot AI', respectively.

92 Each generation consists of multiple round of predictions, and each player is re-
 93 warded a payoff according to the incentive structure (Figure 1C). For simplicity,

we assume that a sufficiently large number of prediction rounds occurs within each generation, allowing us to use the expected payoff value rather than computing the payoffs for every round. We consider two incentive structures from previous work, “feedback” and “niche-expert”, which have been shown to promote near-perfect collective accuracy in the absence of AI [13]. The feedback structure gives higher payoff to individuals who can push the collective prediction toward the true target, regardless of their individual predictive accuracy. [SWP: *maybe we need equations...*] In contrast, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a minority interest. [SWP: *maybe we need equations... but also say that this preserves diversity*] We note collective intelligence fails to emerge under the combination of averaging aggregation with niche-expert structure in the absence of AI [13].

Through time, the group evolves by allowing each individual to imitate peers who have received higher payoffs (Figure 1C). Specifically, after each generation, two individuals are randomly sampled, and one imitates all three strategies (the factor of interest, factor belief, and reliance on AI) of the other with a probability determined by the difference in their expected payoffs. Individual predictions are aggregated into a collective prediction $\hat{Y}$ by simple averaging when using Omniscient AI and by clustering when using Chatbot AI. Collective accuracy is defined as the reduction in prediction error achieved by the collective prediction relative to a baseline prediction α_0 . Hereafter, we refer to the near-perfect collective accuracy as accurate CI.

This model now allows us to explore the impact of AI on CI depending on the mode of AI and the incentive structure. Here, we begin by comparing the emergence of CI with and without AI and subsequently investigate a better incentive structure that can promote CI in the presence of AI.

AI can either promote or prevent the emergence of CI

To understand the impact of AI, we first compare collective dynamics with and without AI. We find that, except when Omniscient AI is used under the feedback structure, a group with AI fails to achieve accurate CI (Fig. 2). However, the effect of AI differs depending on the mode of AI.

When Omniscient AI assists players, the group becomes equally or more accurate than the group without AI. Under the feedback structure, a group achieves accurate CI regardless of the assistance of AI (Fig. 2A). However, under the niche-expert structure, the group without AI is no better than the baseline prediction. Although Omniscient AI cannot help players achieve accurate CI, the group’s prediction with AI is more accurate than both that of the group without AI and the AI itself (Fig. 2B). These results are robust across different levels of AI accuracy and can be shown mathematically by assuming a sufficiently large group of players (Fig. S2).

When Chatbot AI assists players, the group becomes less accurate than the group without AI. Although both incentive structures can achieve accurate CI without AI, the assistance of Chatbot AI hinders group performance and prevents accurate CI

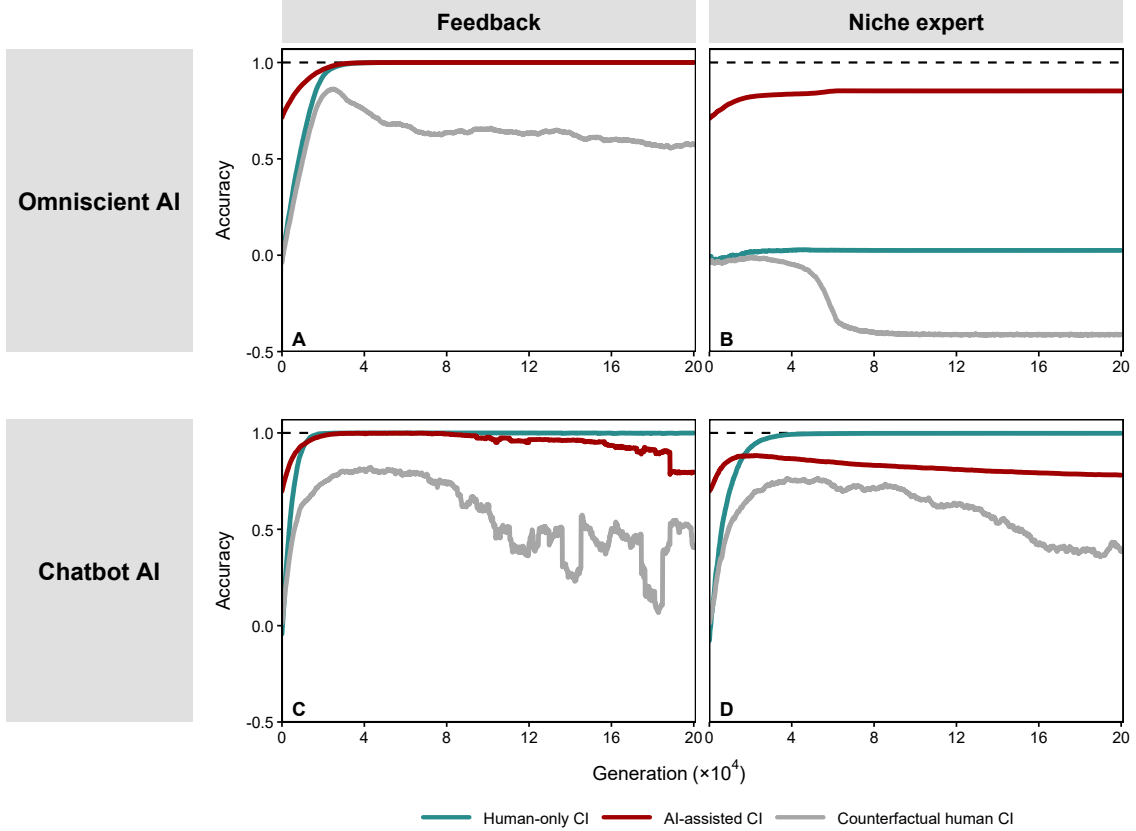


Figure 2: **Need caption.** [SWP: *Need brief description.*]

135 from emerging (Fig. 2C, D). This occurs through different mechanisms depending
 136 on the incentive structure. Under the feedback structure, Chatbot AI leads to a loss
 137 of interest diversity, which eventually makes it impossible for the group to account
 138 for variation in some causal factors. This leads to an increase in collective variance
 139 and stochastic decrease in collective accuracy (Fig. S1). The loss of interest diversity
 140 may arise even without AI after sufficiently many generations in a finite population.
 141 However, Chatbot AI makes the group more vulnerable to the diversity loss. Un-
 142 der the niche-expert structure, the group becomes overly reliant on AI, causing the
 143 collective prediction to inherit the AI’s bias (Fig. S1). This leads to an increase in
 144 collective bias and a decrease in collective accuracy. The niche-expert structure may
 145 also drive reliance on AI toward zero, depending on the relationship between the
 146 true coefficient and AI bias (Fig. S2). In this case, the group completely excludes
 147 AI assistance and builds CI on its own, which substantially delays the emergence of
 148 accurate CI.

149 To investigate whether the group can retain its predictive ability when AI as-
 150 sistance suddenly becomes unavailable at a given generation, we measure counter-
 151 factual collective accuracy. This measure follows the same evolutionary trajectory

as the simulation with AI, but accuracy is computed using the players' unrevised predictions. We find that counterfactual collective accuracy is always lower than the collective accuracy of groups, regardless of the AI mode or incentive structure.

In summary, AI assistance may promote CI when it is omniscient, whereas AI operating with limited information may prevent the emergence of accurate CI. These findings underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

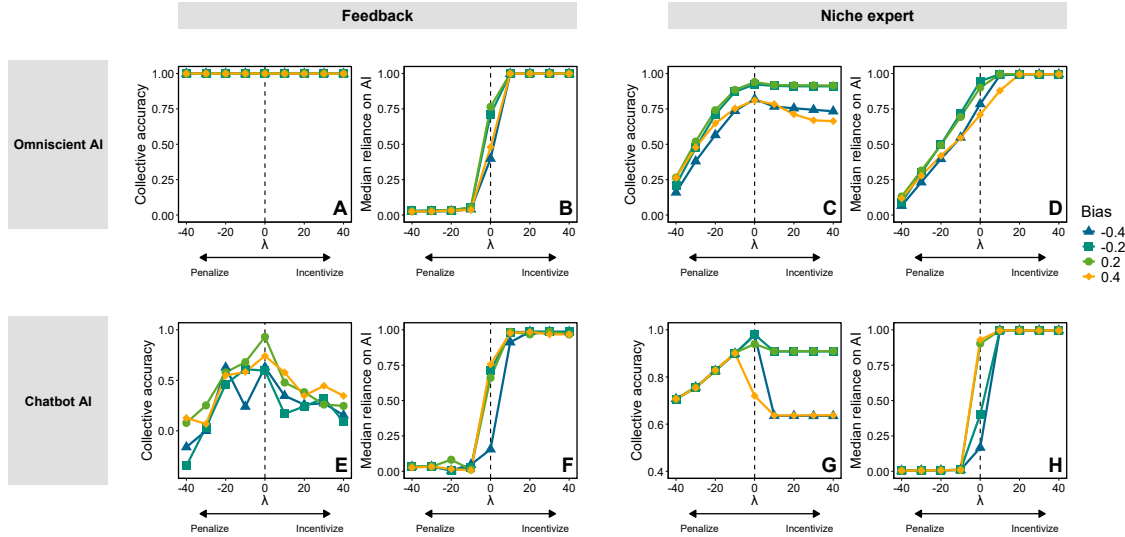


Figure 3: Need caption. [SWP: Need brief description.]

Given the strong influence of AI on collective outcomes, a straightforward intervention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties help emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$).

Directly incentivizing or penalizing AI use further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure (Fig. 3A). This is because incentivizing or penalizing AI usage pushes the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI, respectively (Fig. 3B, D, F, H). In some cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can achieve higher collective accuracy than the original incentive structure. However, in most cases, penalizing AI use imposes strong selection against AI reliance, pushing

173 the group toward lower stationary collective accuracy. Although the group with
 174 Chatbot AI under feedback structure shows stochastic breakdown of CI which doesn't
 175 result in stationary collective accuracy, incentivizing or penalizing AI use further
 176 strengthens the breakdown (Fig. 3E). This result suggests that simply incentivizing
 177 or penalizing the AI use may exacerbate rather than resolving the problem of AI.

178 **Balanced incentives mitigate AI-induced failures of CI**

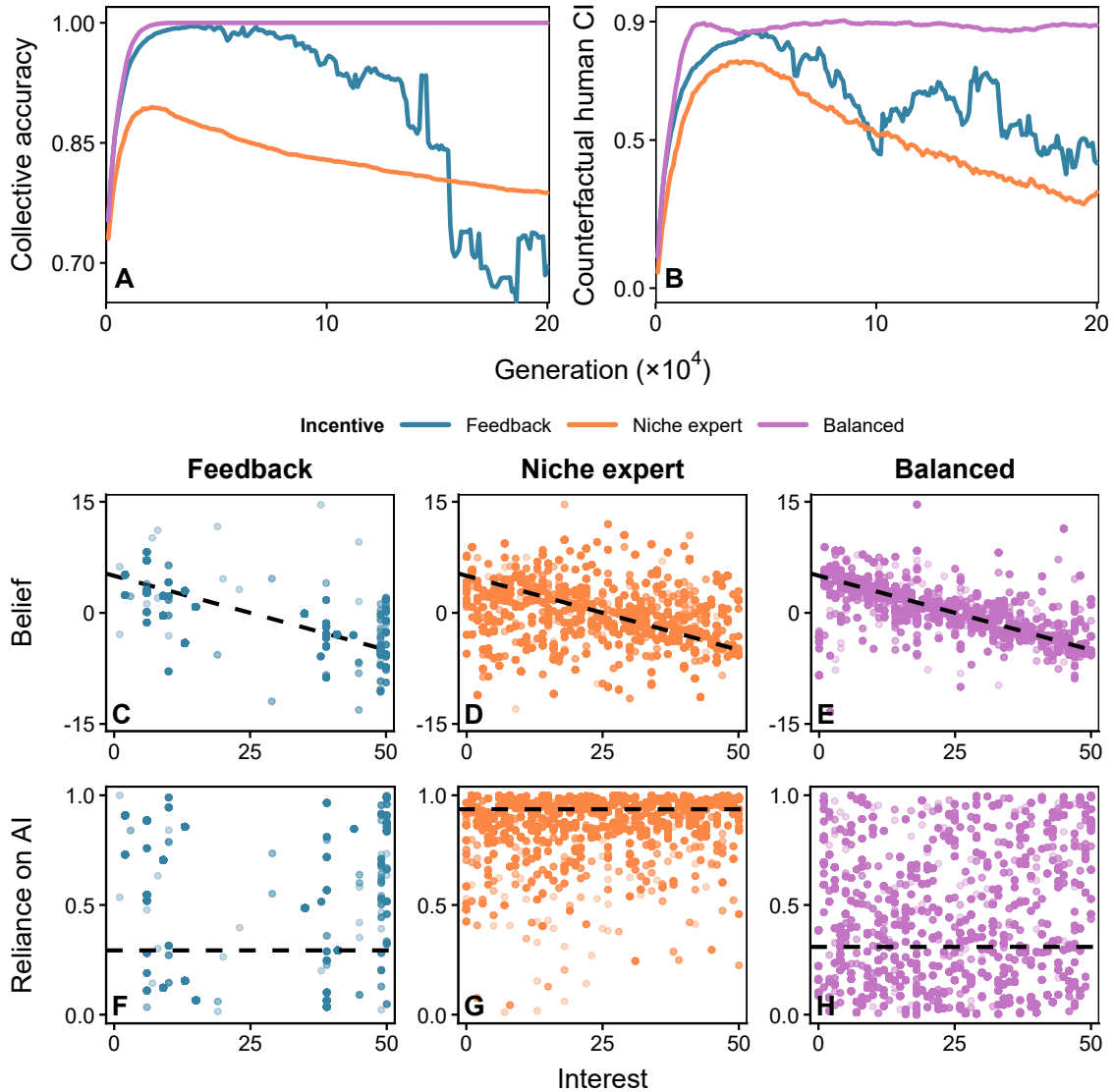


Figure 4: **Need caption.** [SWP: *Need brief description.*]

179 Neither the original incentive structures nor simple interventions targeting AI
 180 use are sufficient to promote CI. What incentive structure, then, would work in the

181 presence of AI? Focusing on the Chatbot AI, in which proposed incentive structures
182 fail to promote accurate CI, we investigate whether a better structure can overcome
183 the risks introduced by AI. Here, we propose a “balanced” structure, defined as the
184 simple average of the feedback and niche expert incentives.

185 Core idea behind the balanced structure is to maintain interest diversity while
186 preventing the collective bias of the group. The feedback structure is prone to in-
187 terest diversity loss, whereas can strongly prevent collective bias by incentivizing
188 players whose predictions lie opposite to the collective prediction. By contrast, the
189 niche-expert structure can maintain interest diversity because it directly incentivizes
190 minority interests. However it cannot prevent collective bias because, within the
191 same factor of interest, individuals are rewarded only according to their individual
192 predictive performance, which homogenize the predictions.

193 By integrating both structure, the balanced structure can promote accurate CI by
194 maintaining both interest diversity and preventing collective bias (Fig. 4, S3). Com-
195 pare to the feedback and niche-expert, players are both evenly distributed through
196 factors, and beliefs are close to the real coefficient (Fig. 4). Moreover, counterfac-
197 tual collective accuracy maintained high compare to both feedback and niche-expert.
198 These patterns are driven by the emergent property of the balanced structure, in
199 which group increases its reliance on AI as the AI becomes more accurate, demon-
200 strating appropriately calibrated AI use.

201 Discussion

202 In this paper, we study how AI and incentive structures affect the emergence of CI.
203 Our model suggests that the effect of AI depends on the type of AI used by the
204 group. AI that can incorporate all relevant causal factors may improve group perfor-
205 mance, whereas AI that relies on users to gather information about the causal factors
206 may impair it. By examining different variants of incentive structures, we further
207 show that direct interventions targeting AI use generally harm group performance.
208 Instead, we suggest balanced structure which can promote CI while also generating
209 appropriately calibrated AI use as an emergent property.

210 Although Omniscient AI can produce group performance that is equal to or bet-
211 ter than that achieved without AI, such AI may be difficult to realize in practice
212 because it needs to be able to observe causal factors independently. Achieving this
213 capability would require hardwares, such as sensors, as well as autonomous data ac-
214 quisition capabilities. For this reason, most AI services currently available are better
215 represented by the Chatbot AI in our model, which highlights the potential risks of
216 AI use for CI.

217 In our model, individuals revise their predictions based on the AI’s assistance. It
218 is therefore unsurprising that AI assistance reduces the diversity of individual pre-
219 dictions, a pattern that has also been observed empirically. However, a reduction in
220 prediction diversity does not necessarily lead to the collapse of CI. The principal risk

221 arises when the incentive structure fails to guide the group toward an appropriate
222 set of strategies. The feedback structure leads to the breakdown of CI because it
223 recruits more players to more important factors, which have larger coefficients. This
224 characteristic creates a risk of interest-diversity loss in finite populations, a situa-
225 tion in which no player remains interested in certain factors because most players
226 shift toward relatively important factors. Without AI, this occurs when initial in-
227 dividual predictions are not sufficiently diverse or when there is no source of novel
228 ideas, such as mutation. Revising individual predictions with AI assistance exac-
229 erbates this problem by pulling individuals’ predictions toward the AI’s prediction,
230 thereby reducing prediction diversity even under the same belief distribution. The
231 niche-expert structure hinders CI because it favors extreme levels of reliance on AI.
232 Because individuals within each interest group are affected only by their own perfor-
233 mance, incentives based on individual performance reach their maximum at either
234 full reliance on AI or no reliance at all. This extreme reliance results in suboptimal
235 collective accuracy or substantially delays the emergence of CI.

236 Because AI already generates substantial benefits as well as costs, organizations
237 increasingly seek to improve collective performance by regulating AI use. Such in-
238 terventions may take direct forms, including AI-use policies or token-based leader-
239 boards, or may operate indirectly through other incentives, such as productivity
240 rewards or the social stigma associated with AI use. However, our model suggests
241 that directly incentivizing or penalizing AI use is unlikely to induce appropriately
242 calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overre-
243 liance, causing groups to propagate AI biases, whereas penalizing AI use could lead
244 to underreliance, preventing groups from using AI even when it performs well in the
245 relevant domain. Our result instead shows that appropriate AI use and accurate CI
246 can be promoted by focusing on the classic properties of groups that have been em-
247 phasized for decades—maintaining diversity and preventing collective bias—rather
248 than on AI use itself.

249 The consistently limited counterfactual collective accuracy observed across AI
250 types and incentive structures suggests that the presence of AI can degrade hu-
251 man group’s own performance, consistent with empirical evidence. This may reduce
252 collective performance when AI assistance becomes unavailable because of resource
253 constraints, system failure, or model collapse, forcing the group to rebuild CI through
254 further adaptation. The higher counterfactual collective accuracy achieved under the
255 balanced structure indicates that an appropriate incentive structure can mitigate the
256 risk of CI breakdown following the loss of AI assistance.

257 Our model and analysis have several limitations. First, the model considers only
258 social learning among humans and treats the influence of AI as unidirectional. In
259 reality, humans can learn from AI, while AI systems can also be further trained
260 on human-generated data. Such bidirectional adaptation may further reduce group
261 performance by increasing the similarity between human and AI predictions and
262 thereby weakening their complementarity. Empirical studies of human–AI collabora-
263 tion likewise suggest that stronger correlations between human and AI predictions

can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random noise. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic calibration. However, AI errors may also exhibit more complex structures, including biased estimates of factor effects and correlations across prediction errors. Further empirical research is therefore needed to characterize the structure of AI bias and error, together with modeling approaches that can incorporate these more complex patterns. Finally, to retain the tractability of established CI models, we assume that the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

To model CI, we start with an abstract model that assumes the prediction target is the result of multiple causal factors [13]. Specifically, the prediction target Y is modeled as a linear combination of M independent factors $(x_1, x_2, \dots, x_m)$ and $M + 1$ coefficients $(\alpha_0, \alpha_1, \dots, \alpha_m)$, where α_0 denotes the global offset (Fig. 1A). The M factors are randomly drawn from normal distributions with mean X_i and standard deviation σ_i in each generation.

Among a population of N individuals, each individual can observe a single factor and make a prediction based on the observed value and their own belief. In particular, individual A 's prediction is modeled as:

$$Y_A = c_A x_{e_A} \quad (2)$$

where c_A represents individual A 's personal belief, e_A represents the factor that A observes (interest), and x_{e_A} represents the realized value of A 's interest. Individuals can also choose not to observe any factor, which represents stubborn individuals who make predictions only based on their own belief. For notational convenience, we assume that stubborn individuals have interest 0, and let x_0 always be 1.

Next, we model AI as a prediction system that has learned the true coefficients with bias and error (Fig. 1B). For each factor $i \in [0, M]$, AI knows the exact coefficient α_i and has a fixed bias b_i , resulting in a prediction of $\alpha_i x_i + b_i$. In terms of human-AI interaction, we model two contrasting modes of AI: the "AI knows all" model and the "AI answers question" model. In the AI knows all model, AI acts as an omniscient system and gives an answer regardless of the individual's interest, accounting for all factors related to the task. Thus, AI predicts as follows:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (3)$$

299 In contrast, the AI answers question model acts as a chatbot and gives an answer
 300 only based on the individual's interest. Thus, given the individual's interest, e_A , AI
 301 predicts as follows:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (4)$$

302 In both models, $\epsilon \sim N(0, \sigma_e^2)$ represents random error arising from the stochastic
 303 generative process of the AI model.

304 AI works as a personalized assistant for each individual, such that each individual
 305 receives assistance by incorporating the AI's prediction. However, even with the same
 306 opportunity to receive AI assistance, reliance on AI can vary substantially across
 307 individuals. To model this, we assume that the revised prediction is a weighted sum
 308 of the individual's own prediction and the AI's prediction by introducing AI belief
 309 β_A as a weighting coefficient. The revised prediction of individual A can be written
 310 as:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (5)$$

311 The revised predictions of individuals are aggregated by two different mechanisms,
 312 depending on the mode of AI. In the AI knows all model, even if an individual can
 313 observe only a single factor, their revised prediction contains information about all
 314 factors through interaction with AI. Therefore, we use simple averaging of revised
 315 predictions as the collective prediction. In contrast, in the AI answers question
 316 model, each individual gives a prediction based only on a single factor even after
 317 revising their prediction with AI assistance. Therefore, we use clustering aggregation
 318 to obtain the collective prediction, which is performed by clustering individuals based
 319 on interest and summing the average values of each cluster.

320 For each generation, each individual is rewarded with a payoff based on the given
 321 incentive structure (Fig. 1C). We adopt two incentive structures, "feedback" and
 322 "niche expert," which were proposed by Wang et al. [13] and promote collective
 323 intelligence when prediction is performed without AI. The feedback incentive gives
 324 higher payoff to individuals who can push the collective prediction toward the true
 325 value, regardless of their individual performance. Given individual A 's revised pre-
 326 diction $Y_A^{revised}$ and the collective prediction $\hat{Y}^{revised}$, it can be expressed as:

$$\pi_A = Y_A^{revised} (Y - \hat{Y}^{revised}) \quad (6)$$

327 The niche expert incentive gives higher payoff to individuals who have both better
 328 individual performance and a minority interest. Specifically, the niche expert payoff
 329 is expressed as the product of the proportion of individuals who observe the same
 330 factor as individual A , ρ_{e_A} , and individual accuracy. Given individual A 's revised
 331 prediction $Y_A^{revised}$, it can be expressed as:

$$\pi_A = -\rho_{e_A} (Y_A^{revised} - Y)^2 \quad (7)$$

These two incentive structures were shown to achieve nearly perfect collective intelligence through social learning (except for the niche expert incentive with simple averaging aggregation) by promoting individual expertise while maintaining collective diversity. Therefore, we use these incentive structures as a baseline to understand how AI affects the emergence of collective intelligence. Moreover, we modify the two baseline incentives in a later section to understand the consequences of direct incentives and penalties for AI usage.

Throughout the simulation, individuals engage in social learning by imitating other individuals who receive higher payoffs. We assume that prediction events occur more frequently than social learning events within the population, such that individual payoff can be computed as an expected payoff. The imitation process is performed for at most one pair in each generation. By sampling two random individuals, A and B , imitation is performed with probability:

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))} \quad (8)$$

where s denotes the strength of social learning. When imitation occurs, A imitates all three strategies of B : interest, belief, and AI belief.

To evaluate and analyze the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, human accuracy, interest diversity, and median AI belief. These metrics capture the accuracy of the revised collective prediction, its bias-variance decomposition, the predictive performance without AI, the diversity of individual interests, and median AI belief in the population (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

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Contributions

Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing - Original Draft Preparation, Writing - Review & Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervision, Writing - Original Draft Preparation, Writing - Review & Editing.

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{revised}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{revised}}]\right)^2$	Squared systematic deviation of the revised collective prediction from the true value.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{revised}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the human-only collective prediction relative to the baseline prediction α_0 .
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median value of reliance on AI β_A in the population.

³⁶⁵ **Competing Interests**

³⁶⁶ The authors declare no competing interests.

³⁶⁷ **Supplementary Text**

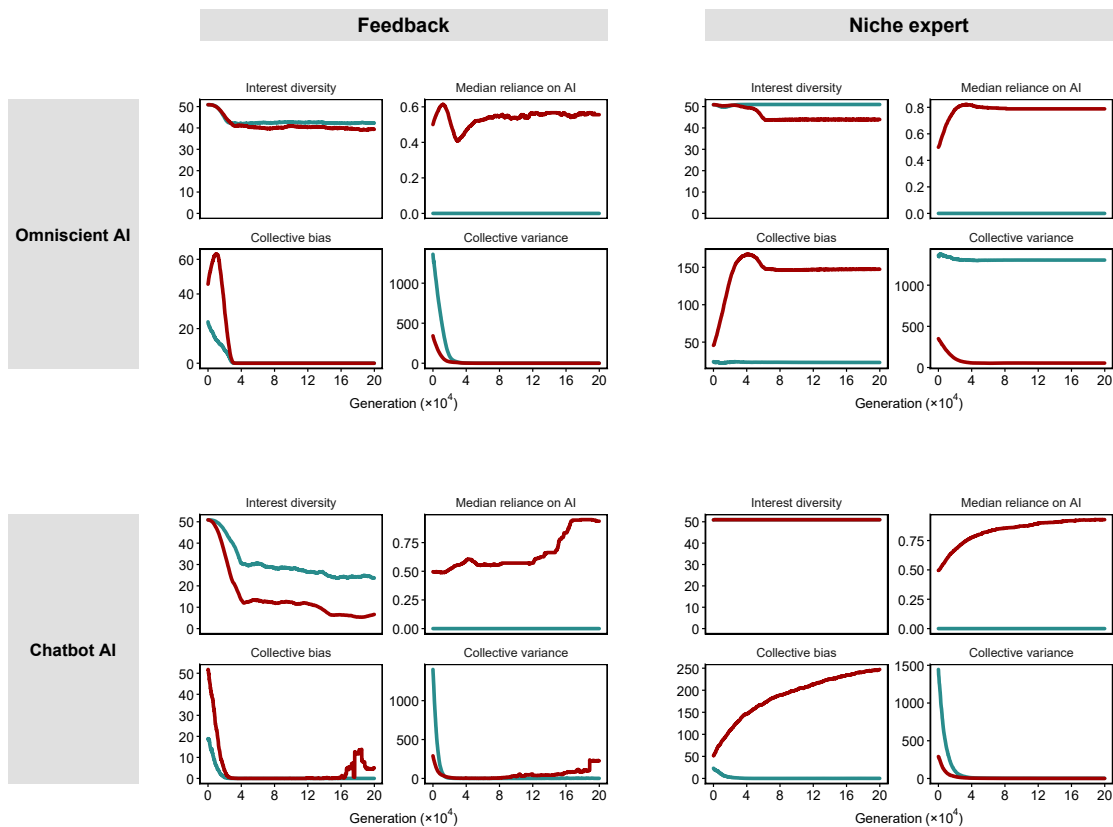


Figure S1: **Caption.** Text.

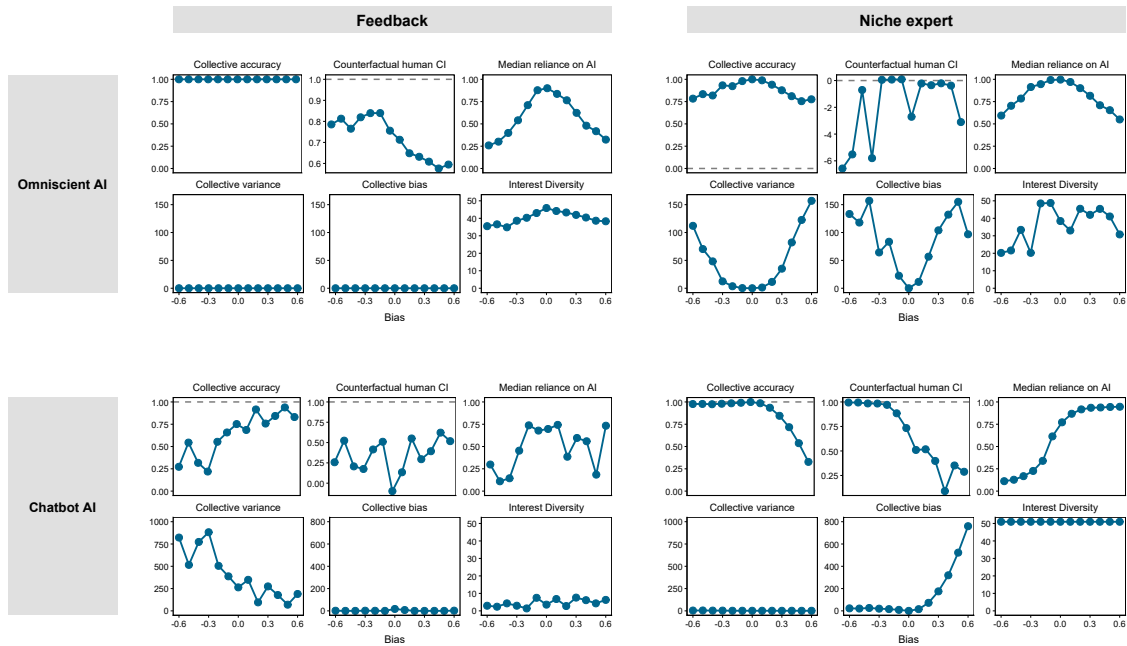


Figure S2: Caption. Text.

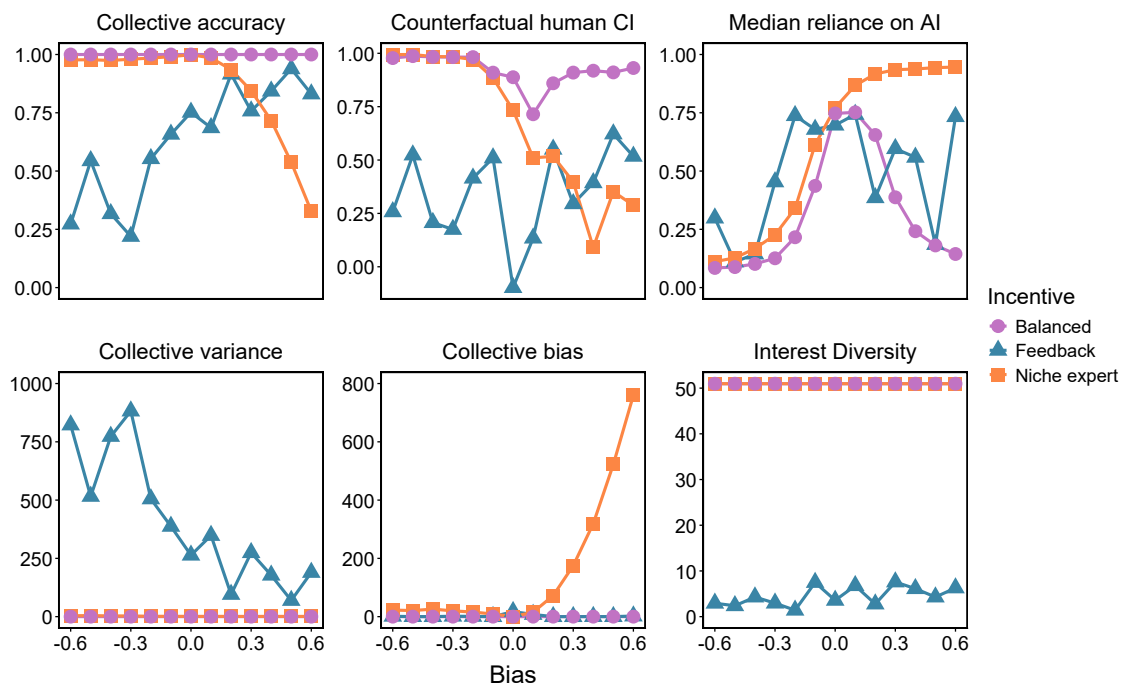


Figure S3: **Caption.** Text.

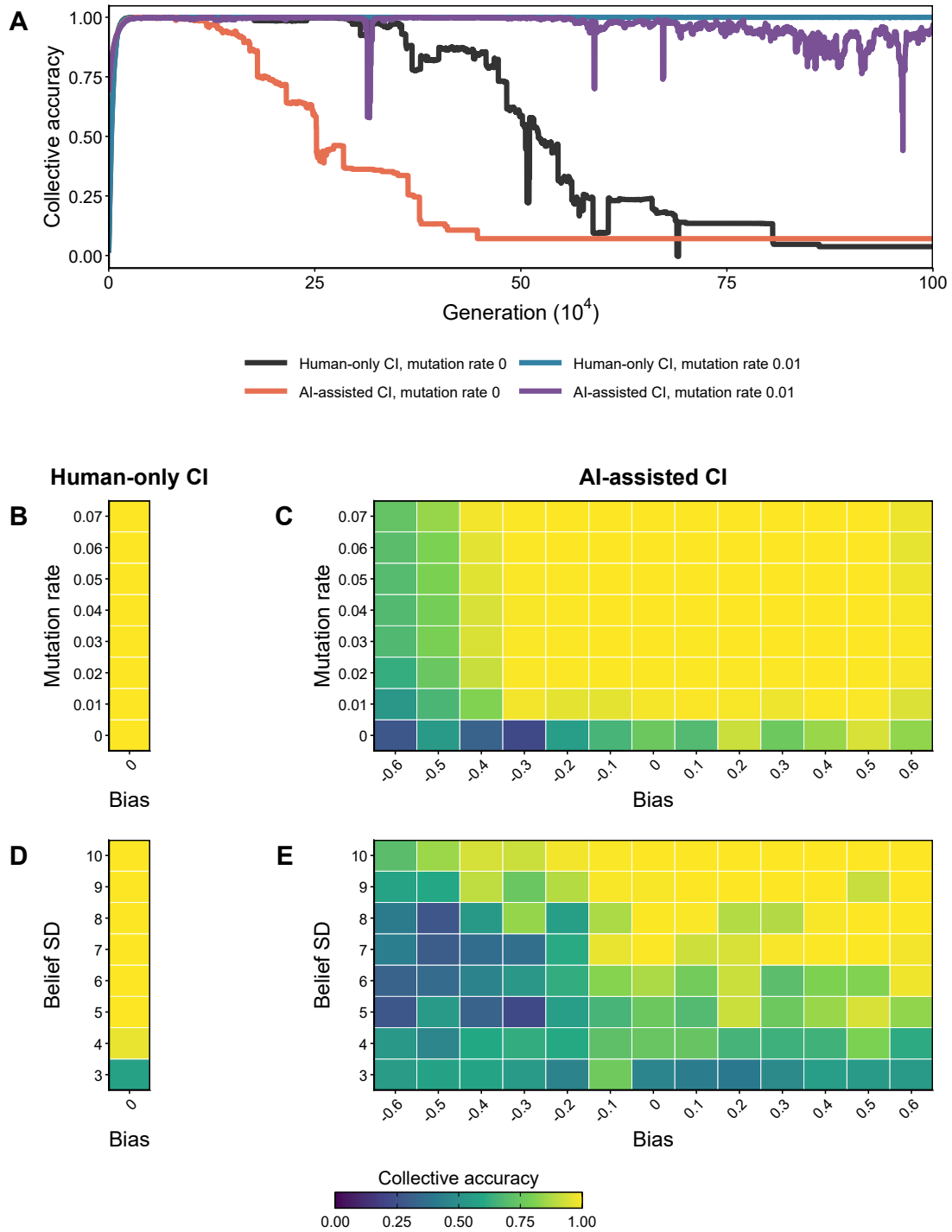


Figure S4: **Caption.** Text.

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